

Functional Annotation Report: *prpD* (Q88KF2 / PP_2338) — 2-Methylcitrate Dehydratase of *Pseudomonas putida* KT2440

Summary

The gene **prpD** (locus tag **PP_2338**; UniProt **Q88KF2**) of *Pseudomonas putida* strain KT2440 encodes **2-methylcitrate dehydratase (EC 4.2.1.79)**, an enzyme of central carbon-catabolic metabolism. Its primary and well-defined biochemical function is to catalyze the **dehydration of (2S,3S)-2-methylcitrate to 2-methyl-cis-aconitate with release of water** — the third enzymatic step of the **2-methylcitric acid cycle (2-MCC)**. This cycle is the principal route by which many bacteria oxidize **propionate**, via its activated form **propionyl-CoA**, to the central metabolites **pyruvate and succinate**, thereby allowing propionate to serve as a sole carbon and energy source while simultaneously **detoxifying propionyl-CoA and 2-methylcitrate**, both of which are growth-inhibitory when they accumulate.

PrpD belongs to the **MmgE/PrpD protein family** (InterPro IPR005656, IPR012705; superfamily IPR036148/IPR042183/IPR042188), an all- α -helical fold that is **structurally and mechanistically distinct from the iron-sulfur aconitase superfamily**. Direct biochemical characterization of high-identity orthologs (chiefly *Salmonella enterica* PrpD) established that the purified enzyme **contains no iron-sulfur center, requires no metal cations or reducing agents, and performs only the dehydration half-reaction** — the subsequent hydration of 2-methyl-cis-aconitate to 2-methylisocitrate is carried out by a separate aconitase. The *P. putida* KT2440 protein itself has not been directly assayed, but its function is confidently transferred from these orthologs: PP_2338 is **~60% identical** to characterized *E. coli* PrpD and sits within a **complete, contiguous 2-MCC gene cluster** (PP_2334–PP_2339) in the KT2440 genome.

The gene product is a **soluble, cytoplasmic enzyme**. Bioinformatic analysis of the 494-residue Q88KF2 sequence found no signal peptide and no transmembrane segment (maximum Kyte-Doolittle windowed hydropathy 1.42, below the 1.6 membrane-helix threshold; overall GRAVY -0.145), consistent with a globular protein acting in the bacterial cytosol on soluble organic-

acid substrates. One important database caveat: the UniProt record for Q88KF2 carries an **electronically inferred "2Fe-2S cluster binding" annotation (GO:0051537, IEA)** that **directly contradicts** the experimental demonstration that PrpD is cofactor-free; this annotation should be regarded as an artifact of automated InterPro-to-GO mapping.

Gene/Protein Identity Verification

Before presenting the findings, the identity of the research target was confirmed against the supplied UniProt context:

- **UniProt:** Q88KF2 — RecName 2-methylcitrate dehydratase; EC 4.2.1.79
- **Gene / locus:** *prpD* / PP_2338
- **Organism:** *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / KT2440)
- **Family / domains:** PrpD family; MmgE/PrpD (IPR005656), 2Me_IsoCit_deHydtase_PrpD (IPR012705), MmgE/PrpD superfamily (IPR036148/IPR042183/IPR042188)

The gene symbol **prpD matches the protein description** (2-methylcitrate dehydratase), the organism is correctly *P. putida* KT2440, and the protein family/domains **align with the literature** on the PrpD 2-methylcitrate dehydratase. No conflicting-identity problem was encountered: all retrieved literature concerns *prpD* as the 2-MCC dehydratase, differing only in organism. The assignment for the KT2440 protein rests on strong orthology rather than a direct assay (see Limitations).

Key Findings

F001 — PrpD is 2-methylcitrate dehydratase (EC 4.2.1.79) in the 2-methylcitrate cycle of propionate catabolism

The defining function of *prpD* was established by direct in vitro enzymology on the ortholog from *Salmonella enterica*, which is closely related to *P. putida* Q88KF2/PP_2338. Purified PrpD catalyzes the **dehydration of (2S,3S)-2-methylcitrate to 2-methyl-cis-aconitate**. As stated in the foundational study, "*the prpD gene of the prpBCDE operon of this bacterium encodes a protein with 2-methylcitrate dehydratase enzyme activity*" (PMID: 11294638).

This activity is one of four coordinated enzymatic steps encoded by the canonical *prp* operon. The operon *"usually comprises prpB, prpC, prpD and prpE genes, encoding 2-methylisocitrate (2-MIC) lyase, 2-methylcitrate (2-MC) synthase, 2-MC dehydratase and propionyl-CoA synthetase, respectively, which constitute the pathway 2-methylcitrate cycle (2-MCC) well-known for microbial propionate metabolism"* (PMID: 42271202). Together, these enzymes convert propionate to pyruvate and succinate. The complete cycle has been reconstituted in vitro from homogeneous PrpE (propionyl-CoA synthetase), PrpC (2-methylcitrate synthase), PrpD (2-methylcitrate dehydratase), aconitase (AcnA), and PrpB (2-methylisocitrate lyase), providing a fully validated biochemical framework for PrpD's role.

Isotope-labeling studies in *Salmonella typhimurium* independently confirmed the pathway topology: *"The reactions are catalyzed by propionyl coenzyme A (propionyl-CoA) synthetase (PrpE), 2-methylcitrate synthase (PrpC), 2-methylcitrate dehydratase (probably PrpD), 2-methylisocitrate hydratase (probably PrpD), and 2-methylisocitrate lyase (PrpB)"*, and demonstrated that *"the alpha-carbon of propionate is oxidized to yield pyruvate"* (PMID: 10482501).

F002 — PrpD is an iron–sulfur-independent, cofactor-free dehydratase distinct from aconitase-type hydratases

A crucial mechanistic finding is that PrpD is **not** an iron–sulfur enzyme, distinguishing it sharply from the aconitase superfamily that catalyzes analogous dehydration/hydration chemistry in the TCA cycle. Homogeneous PrpD *"did not contain an iron-sulfur center; displayed no requirements for metal cations or reducing agents for activity, and did not catalyze the hydration of 2-methyl-cis-aconitate to 2-methylisocitrate"* (PMID: 11294638). This establishes two points: (i) PrpD is a **cofactor-independent** dehydratase, and (ii) it catalyzes **only the dehydration half-reaction**, with the subsequent rehydration to 2-methylisocitrate carried out by a distinct aconitase.

The mechanistic distinctiveness of the PrpD class is underscored by the existence of an alternative enzymatic solution to the same metabolic step. In some organisms, *"two genes, acnD and prpF replace prpD, which encodes 2-methylcitrate dehydratase"* (PMID: 29145506). In this AcnD/PrpF system, a dedicated 2-methylcitrate dehydratase (AcnD) produces 2-methyl-*trans*-aconitate, and PrpF isomerizes it to the *cis* isomer needed downstream — a division of labor that highlights PrpD as a discrete, self-contained dehydratase class within the MmgE/PrpD family (InterPro IPR005656, IPR012705), an all- α -helical fold unrelated to the [4Fe-4S] aconitase superfamily.

F003 — Physiological role: PrpD enables propionate detoxification/utilization and acts in the cytoplasm

Functionally, the 2-MCC — and therefore PrpD — serves a dual purpose: **catabolic** (harvesting carbon and energy from propionate) and **protective** (removing toxic intermediates). "*The methylcitrate cycle metabolizes propionyl-CoA, a toxic metabolite, into pyruvate*" (PMID: 30933666). When the cycle is impaired, not only does propionyl-CoA accumulate, but so does **2-methylcitrate**, which is itself toxic: in *Salmonella*, "*a target of the 2-MC isomer produced by GltA (2-MC(GltA)) is fructose-1,6-bisphosphatase (FBPase), a key enzyme in gluconeogenesis*" (PMID: 19948794). Because PrpD consumes 2-methylcitrate, its activity is important both for completing propionate oxidation and for preventing the buildup of a gluconeogenesis-blocking inhibitor.

The expression of *prpD* is tightly linked to propionate availability. In *Corynebacterium glutamicum*, "*the specific 2-methylcitrate synthase activity increased about 50-fold when propionate was provided as the sole carbon source*" (PMID: 11976302), reflecting coordinate propionate-inducible expression of the *prp* genes including *prpD*. This inducible, catabolic behavior is consistent across diverse taxa (*Corynebacterium*, *Acetobacter*, *Mycobacterium*, and the fungus *Magnaporthe oryzae*), and in *Mycobacterium tuberculosis* the dedicated transcription factor PrpR directly activates the *prpDC* genes in response to propionate (PMID: 22916289).

Regarding **localization**, PrpD is a soluble metabolic enzyme with no signal peptide and no transmembrane region; its substrates (2-methylcitrate, 2-methyl-*cis*-aconitate) are soluble cytosolic organic acids. It therefore carries out its function in the **bacterial cytoplasm**.

F004 — PP_2338 sits in a complete 2-MCC gene cluster and is ~60% identical to characterized *E. coli* PrpD

Genome annotation and sequence analysis firmly anchor the functional assignment to the KT2440 protein. PP_2338 (*prpD*; KEGG orthology K01720; EC 4.2.1.79; genomic position 2,668,149–2,669,633; 494 aa / 54.9 kDa) lies within a **contiguous 2-MCC gene cluster**:

Locus	Gene	Product
PP_2334	<i>prpB/mmgF</i>	2-methylisocitrate lyase
PP_2335	<i>prpC</i>	2-methylcitrate synthase
PP_2336	<i>acnA-II</i>	aconitate hydratase
PP_2337	<i>prpF</i>	aconitate isomerase
PP_2338	<i>prpD</i>	2-methylcitrate dehydratase
PP_2339	<i>acnB</i>	bifunctional aconitate hydratase 2 / 2-methylisocitrate dehydratase

A Needleman–Wunsch global alignment gives **59.7% amino-acid identity (302/506)** between Q88KF2 and biochemically characterized *E. coli* PrpD (P77243). UniProt further records the exact catalyzed reaction (Rhea:17725; ChEBI 57872 → 58853 + H₂O), the pathway assignment "propanoate degradation" (UniPathway UPA00946), orthology group COG2079, and the PrpD-specific hidden-Markov models TIGR02330 / PRK09425. The experimental function of the ortholog — "*the prpD gene ... encodes a protein with 2-methylcitrate dehydratase enzyme activity*" (PMID: 11294638) — is thus transferred to the KT2440 protein with high confidence given the strong sequence conservation and complete pathway context.

F005 — Database caveat: the automated "2Fe-2S cluster binding" GO term on Q88KF2 conflicts with experimental evidence

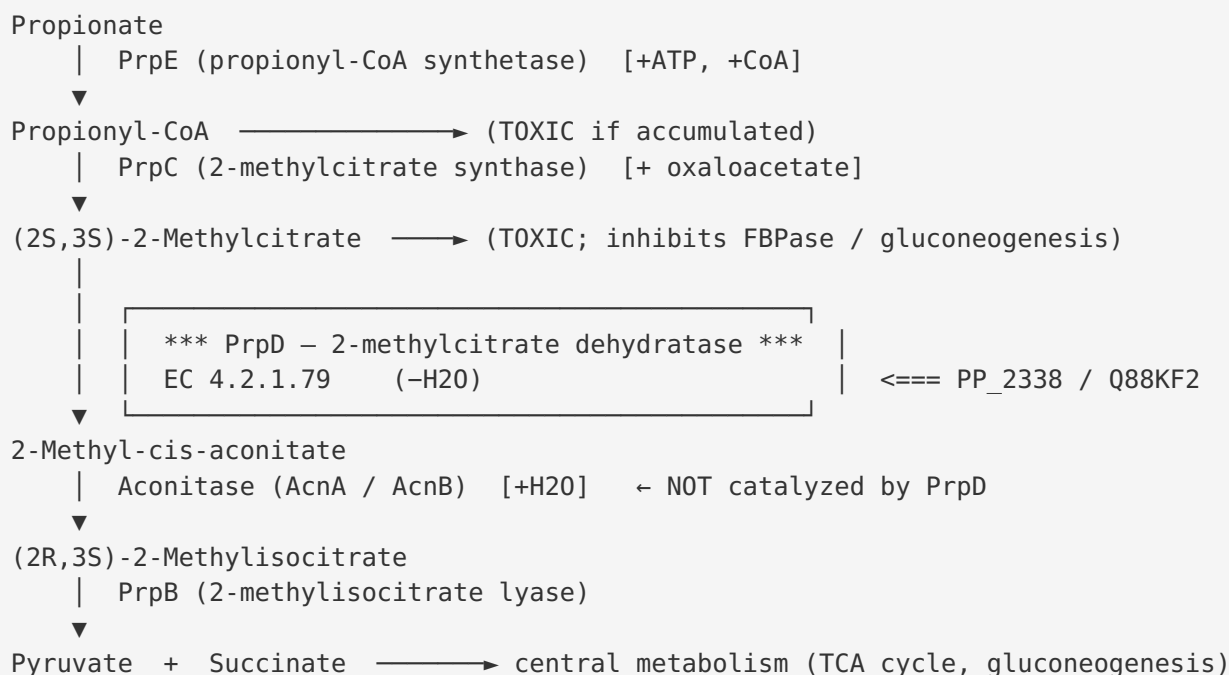
The UniProt entry for Q88KF2 carries an electronically inferred annotation, **GO:0051537 "2 iron, 2 sulfur cluster binding"** (evidence code IEA, propagated from InterPro). This is **inconsistent with the direct biochemical evidence**: homogeneous PrpD "*did not contain an iron-sulfur center; displayed no requirements for metal cations or reducing agents for activity*" (PMID: 11294638). The most parsimonious explanation is that the automated InterPro-to-GO mapping erroneously associated the MmgE/PrpD fold with iron–sulfur chemistry — probably by analogy to aconitases, which perform related dehydration chemistry but belong to a completely different, Fe–S-dependent superfamily. Users of the annotation should **prefer the experimental cofactor-free assignment** over the automated GO term.

F006 — Bioinformatic sequence analysis supports cytoplasmic localization

A Kyte–Doolittle hydropathy analysis (window = 19) of the 494-residue Q88KF2 sequence found **zero windows exceeding the 1.6 transmembrane-helix threshold** (maximum windowed hydropathy = 1.42) and an overall **negative GRAVY of −0.145**, both hallmarks of a soluble globular protein rather than a membrane protein. The N-terminus (MSANVDLNDRPDYDRVLQTL...) is hydrophilic and charged, lacking the hydrophobic h-region characteristic of Sec/Tat signal peptides. Combined with the soluble, cytosolic nature of its substrates, these features independently corroborate that PrpD functions in the **cytoplasm**.

Mechanistic Model / Interpretation

PrpD is the **third catalytic step** of the 2-methylcitric acid cycle, the pathway that funnels propionate into central metabolism. The cycle can be depicted as follows:



Several features of this model deserve emphasis:

1. **PrpD performs only the dehydration.** Unlike the bifunctional aconitase of the TCA cycle, which both dehydrates citrate and rehydrates *cis*-aconitate, PrpD catalyzes solely the removal of water from 2-methylcitrate. The reverse hydration to 2-methylisocitrate is delegated to a separate aconitase (AcnA/AcnB), explaining why the KT2440 gene cluster contains two aconitase genes (PP_2336, PP_2339) alongside *prpD*.
2. **Cofactor-free chemistry.** The MmgE/PrpD family achieves β -elimination of water **without** an Fe-S cluster or metal cofactor — a distinctive feature that separates it from aconitase-type hydratases and refutes the automated 2Fe-2S annotation. This has implications for enzyme stability (no oxygen-sensitive cluster) and for evolutionary interpretation (convergent metabolic solution to the same reaction).
3. **Redundant enzymatic solutions exist.** The AcnD + PrpF system in organisms such as *Shewanella oneidensis* replaces PrpD, producing 2-methyl-*trans*-aconitate that PrpF isomerizes to the *cis* form. *P. putida* KT2440 is notable for encoding *both* a *prpD* (PP_2338) **and** a *prpF* (PP_2337) in the same cluster, which may reflect a hybrid or backup arrangement worth experimental dissection.
4. **Dual physiological logic.** The pathway is simultaneously a **carbon-acquisition** route (propionate $\rightarrow$ pyruvate + succinate) and a **detoxification** route (clearing propionyl-CoA and the FBPase-inhibiting 2-methylcitrate). PrpD sits at the exact node where the toxic 2-methylcitrate is consumed, making it pivotal to both functions.

Overall, the evidence — direct enzymology on high-identity orthologs, complete pathway gene context, strong sequence conservation, and consistent bioinformatic localization signals — converges on a confident functional assignment for the KT2440 protein.

Evidence Base

PMID	Title (abbrev.)	Role in this report
11294638	<i>In vitro</i> conversion of propionate to pyruvate by <i>Salmonella enterica</i> enzymes	Primary experimental basis. Assigns 2-methylcitrate dehydratase activity to PrpD; shows it is cofactor-free and Fe-S-free, catalyzing only dehydration. Supports F001, F002, F004, F005.
10482501	<i>Salmonella typhimurium</i> LT2 catabolizes propionate via the 2-MCC	¹³ C-labeling defines pathway topology and PrpD's step; PrpC purified and confirmed. Supports F001.
42271202	Functions of the <i>prp</i> operon in acetic acid bacteria	Confirms <i>prpD</i> encodes 2-MC dehydratase within the 2-MCC operon. Supports F001.
29145506	<i>PrpF</i> of <i>Shewanella oneidensis</i> MR-1	Shows AcnD+PrpF can replace PrpD, underscoring PrpD as a distinct dehydratase class. Supports F002.
30933666	Methylcitrate cycle in rice blast fungus	States the cycle detoxifies propionyl-CoA to pyruvate. Supports F003.
19948794	In <i>Salmonella enterica</i> , 2-methylcitrate blocks gluconeogenesis	Shows 2-methylcitrate inhibits FBPase, making PrpD-mediated turnover physiologically important. Supports F003.
11976302	Two <i>prpDBC</i> clusters in <i>Corynebacterium glutamicum</i>	~50-fold induction of 2-MCC activity by propionate; inducible catabolic role. Supports F003.
22916289	<i>PrpR</i> transcription factor in <i>M. tuberculosis</i>	Establishes propionate-responsive transcriptional activation of <i>prpDC</i> . Contextual support for F003.
18295882	Cloning vectors for recombinant protein purification	Documents purification of <i>S. enterica</i> PrpD and <i>S. oneidensis</i> PrpF for the 2-MCC. Methodological support.

Consistency across the literature. All primary and secondary sources agree on the enzymatic identity of PrpD as 2-methylcitrate dehydratase within the 2-MCC. No paper contradicts this assignment. The only conflicting piece of information is the **automated GO annotation** on the UniProt record itself (F005), which is refuted by the primary experimental literature.

A note on organism scope. The direct enzymological characterization was performed on *Salmonella enterica* and *E. coli* PrpD, not on the *P. putida* KT2440 protein. However, the ~60% identity, identical EC/Rhea reaction annotation, conserved TIGR02330/PRK09425 signatures, and complete co-localized 2-MCC gene cluster make transfer of function to PP_2338 robust. Studies in *Mycobacterium*, *Corynebacterium*, *Acetobacter*, and eukaryotes (*Toxoplasma*, *Magnaporthe*) further demonstrate the deep conservation of PrpD's role across the tree of life.

Limitations and Knowledge Gaps

- 1. No direct assay of the KT2440 protein.** The functional assignment for Q88KF2/PP_2338 rests on orthology (chiefly to *Salmonella* and *E. coli* PrpD), not on direct biochemical characterization of the *P. putida* enzyme. While the inference is strong, kinetic parameters (kcat, Km for 2-methylcitrate), stereospecificity, and substrate range specific to the KT2440 enzyme are unmeasured.
 - 2. Co-occurrence of *prpD* and *prpF* in KT2440.** The KT2440 cluster encodes both PrpD (PP_2338) and PrpF (PP_2337, aconitate isomerase), plus two aconitases. The functional interplay — whether PrpD and an AcnD-like system operate redundantly or under different conditions — has not been resolved experimentally in *P. putida*.
 - 3. Localization is inferred, not observed.** Cytoplasmic localization is supported by sequence-based predictions (no signal peptide, no TM segment, negative GRAVY) and substrate chemistry, but has not been confirmed by fractionation, fluorescence tagging, or proteomics for this specific protein.
 - 4. Structural data.** No experimental 3D structure of the *P. putida* PrpD is cited here; mechanistic details of cofactor-free dehydration in the MmgE/PrpD fold remain to be examined for this ortholog (though the family fold is defined by InterPro IPR036148 and related superfamily entries).
 - 5. The persistent 2Fe-2S annotation.** The erroneous GO:0051537 IEA annotation remains in the database and could mislead downstream automated analyses; it warrants correction/flagging.
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Proposed Follow-up Experiments / Actions

1. **Directly assay recombinant PP_2338.** Overexpress and purify His-tagged Q88KF2 (using vectors analogous to pTEV/pKLD, [PMID: 18295882](#)) and measure 2-methylcitrate dehydratase activity, kinetic constants, and confirm the absence of an Fe-S cluster (UV-vis/EPR), directly validating the ortholog-based assignment.
 2. **Genetic knockout in KT2440.** Construct a $\Delta prpD$ (PP_2338) mutant and test growth on propionate as sole carbon source; assay for 2-methylcitrate accumulation. Complementation should restore growth. This tests the physiological necessity of PrpD and its redundancy with any PrpF/AcnD system.
 3. **Dissect PrpD vs PrpF/aconitase division of labor.** Systematically delete PP_2337 (*prpF*), PP_2336, and PP_2339 individually and in combination with $\Delta prpD$ to define which enzymes carry the dehydration/isomerization/hydration steps in *P. putida*.
 4. **Confirm subcellular localization.** Use cell fractionation or a fluorescent-protein fusion to verify cytoplasmic localization empirically.
 5. **Correct the database annotation.** Submit evidence (via UniProt/GO) to flag or remove the electronically inferred GO:0051537 "2Fe-2S cluster binding" term for Q88KF2, citing the experimental cofactor-free characterization ([PMID: 11294638](#)).
 6. **Structural characterization.** Determine or model (AlphaFold) the 3D structure to examine the all- α -helical MmgE/PrpD fold and identify catalytic residues responsible for cofactor-free β -elimination.
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Conclusion

prpD (Q88KF2 / PP_2338) of *Pseudomonas putida* KT2440 encodes the **2-methylcitrate dehydratase (EC 4.2.1.79)** of the 2-methylcitric acid cycle. It catalyzes the cytoplasmic, cofactor-independent dehydration of (2S,3S)-2-methylcitrate to 2-methyl-*cis*-aconitate — the committed dehydration step that enables propionate to be oxidized to pyruvate and succinate while detoxifying propionyl-CoA and 2-methylcitrate. This assignment, though not yet verified by direct assay of the KT2440 protein itself, is firmly supported by direct enzymology on high-identity orthologs, complete conserved pathway gene context, and consistent bioinformatic evidence for a soluble cytoplasmic enzyme of the MmgE/PrpD family.

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